

Notes on two walks with noncollinear drifts

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We use the notation:

- polar form of the walk: $S_k^{(j)} = \|S_k^{(j)}\| e_{\varphi_k^{(j)}}$; then

$$\langle S_k^{(j)}, e_\theta \rangle = \|S_k^{(j)}\| \cos(\varphi_k^{(j)} - \theta)$$

- for the index k where the max of $\langle S_k^{(j)}, e_\theta \rangle$ is achieved, we write $k^*(\theta, n, j) := k$

1 Lemmas

By the SLLN, all points $S_k^{(1)}$ for $k \geq n$ where $n = n(\omega, \delta)$ is a random and a.s. finite $\in \mathbb{N}$ are contained in the cone

$$\text{Co}^{(1)} = \left\{ re_\theta : r \geq 0, \theta \in \left\langle \frac{\pi}{4} - \delta, \frac{\pi}{4} + \delta \right\rangle \right\}$$

and likewise for the second walk.

Lemma 1. *A.s. for large enough n it holds that*

$$S_{k^*(\theta, n, 1)}^{(1)} \in \text{Co}^{(1)}, \quad \forall \theta \in \left\langle -\frac{\pi}{4} + 2\delta, \frac{3\pi}{4} - 2\delta \right\rangle,$$

Proof. Write

$$\max_{1 \leq m \leq n} \langle S_m^{(1)}, e_\theta \rangle = \max \left\{ \max_{S_m^{(1)} \notin \text{Co}^{(1)}} \langle S_m^{(1)}, e_\theta \rangle, \max_{S_m^{(1)} \in \text{Co}^{(1)}} \langle S_m^{(1)}, e_\theta \rangle \right\} \quad (1)$$

The first part of (1) will a.s. at some point remain constant, and can therefore be bounded by the maximum of norms (making it independent of θ as well as n).

For the second part note first

$$\langle S_m^{(1)}, e_\theta \rangle = \|S_m^{(1)}\| \cos(\theta - \varphi_m^{(1)}).$$

When $S_m^{(1)} \in \text{Co}^{(1)}$ we have $\cos(\theta - \varphi_m^{(1)}) > 0$ and $\|S_m^{(1)}\| \xrightarrow{\text{a.s.}} \infty$ meaning that the maximum in (1) is a.s. always generated by the second part for large n .

Moreover,

$$\cos(\theta - \varphi_m^{(1)}) > \cos\left(-\frac{\pi}{2} + \delta\right) \quad (2)$$

which is a strictly positive constant, meaning that we can choose an n which is “good” for all such θ simultaneously. $\square$

Note 2. When

$$\theta \in \left\langle -\frac{\pi}{4} + \delta, \frac{\pi}{4} - \delta \right\rangle,$$

we cannot repeat the same argument, as the RHS in (2) is 0.

Note 3. For any fixed $\theta \in \langle -\pi/4, 3\pi/4 \rangle$ we can retroactively choose a small enough δ to claim $S_{k^*(\theta, n, 1)} \in \text{Co}^{(1)}$ a.s. for large enough n , but we cannot do so for all θ simultaneously.

Note 4. Whenever θ is instead such that $\cos(\theta - \varphi_m^{(1)}) < 0$ inside the cone, the maximum is generated either by 0 or by a point outside the cone, meaning that the maximum a.s. eventually “freezes”. To make the analogous claim to lemma 1

$$\theta \in \left\langle \frac{3\pi}{4} + \delta, \frac{7\pi}{4} - \delta \equiv -\frac{\pi}{4} - \delta \right\rangle$$

is enough, since the $\langle S_n^{(1)}, e_\theta \rangle$ become negative for large enough n , and the magnitude of the cosine (and therefore θ itself) plays no part here.

2 Main part

Consider the limiting behaviour of the integral

$$\frac{1}{n^{3/2}} \int_0^\pi [M_{n,2}(\theta)^2 - M_{n,1}(\theta)^2] \cdot 1_{\{M_{n,1}(\theta) < M_{n,2}(\theta)\}} d\theta.$$

For $\theta \in [\varepsilon, \pi/4 + \delta]$. We can bound the indicator uniformly in θ using

$$\{M_{n,1}(\theta) < M_{n,2}(\theta)\} \subseteq \left\{ \min_{\theta \in [\varepsilon, \pi/4 + \delta]} M_{n,1}(\theta) < \max_{\theta \in [\varepsilon, \pi/4 + \delta]} M_{n,2}(\theta) \right\}, \quad \forall \theta \in [\varepsilon, \pi/4 + \delta].$$

For n large enough in the sense of lemma 1 for both walks we claim:

$$\min_{\theta \in [\varepsilon, \pi/4 + \delta]} M_{n,1}(\theta) = M_{n,1}(\varepsilon), \quad \max_{\theta \in [\varepsilon, \pi/4 + \delta]} M_{n,2}(\theta) = M_{n,2}(\varepsilon).$$

We give the proof for the first walk. Assuming the contrary, there is a $\theta > \varepsilon$ and indices j, k generating maxima for θ, ε respectively such that

$$\langle S_j^{(1)}, e_\theta \rangle < \langle S_k^{(1)}, e_\varepsilon \rangle.$$

But then

$$\langle S_k^{(1)}, e_\theta \rangle \leq \langle S_j^{(1)}, e_\theta \rangle < \langle S_k^{(1)}, e_\varepsilon \rangle$$

implying

$$\cos(\varphi_k^{(1)} - \theta) < \cos(\varphi_k^{(1)} - \varepsilon)$$

yielding a contradiction since $\varphi_k^{(1)}$ corresponds to the cone $\text{Co}^{(1)}$.

Since by the SLLN

$$1_{\{M_{n,1}(\varepsilon) < M_{n,2}(\varepsilon)\}} \xrightarrow{\text{a.s.}} 0, \quad \forall \varepsilon > 0,$$

this proves convergence a.s. to 0. The proof for the second walk is analogous.¹

¹note that to apply lemma 1 we'd need the upper endpoint to be $\pi/4 - 2\delta$ instead. However, for this proof to work we only need the result at the fixed $\theta = \varepsilon$. The upper endpoint $\pi/4 + \delta$ is chosen for the sake of the next part.

For $\theta \in [0, \varepsilon]$. Note that above we've also shown that

$$\theta \mapsto M_{n,1}(\theta)$$

is increasing on the domain $[0, \varepsilon]$ (a.s. for a large-enough random n). We prove L^1 -convergence.

For $\theta \in \langle \pi/4 + \delta, \pi \rangle$ Here $M_{n,2}(\theta)$ will a.s. freeze (at the same time for all θ), so simply

$$M_{n,2}(\theta)^2 - M_{n,1}(\theta)^2 \leq M_{n,2}(\theta)^2 \leq \max_{S_m^{(2)} \notin \text{Co}(2)} \|S_m^{(2)}\|^2, \quad \text{on } \{M_{n,1}(\theta) < M_{n,2}(\theta)\}$$

makes a.s. convergence to 0 under $n^{-3/2}$ scaling obvious.

For $\theta \in \langle -\pi, 0 \rangle$, the situation is symmetrical to the previous 3 points.

3 Derivatives

Writing S_n for a generic walk, Pythagoras' theorem gives

$$\|S_{k^*(\theta,n)}\|^2 = M_n(\theta)^2 + M'_n(\theta)^2,$$

meaning that

$$M'_{n,2}(\theta)^2 - M'_{n,1}(\theta)^2 = \left[\|S_{k^*(\theta,n,2)}^{(2)}\|^2 - \|S_{k^*(\theta,n,1)}^{(1)}\|^2 \right] - [M_{n,2}(\theta)^2 - M_{n,1}(\theta)^2]$$

The norms can be bounded using M_n . First note

$$\|S_{k^*(\theta,n,1)}^{(1)}\| = \frac{M_{n,1}(\theta)}{\cos(\theta - \varphi_{k^*(\theta,n,1)}^{(1)})}$$

For example when $\theta \in \langle 0, \varepsilon \rangle$ we can guarantee

$$\varphi_{k^*(\theta,n,1)}^{(1)} - \theta \in \left\langle \frac{\pi}{4} - \delta - \varepsilon, \frac{\pi}{4} + \delta \right\rangle$$

(again using lemma 1). Doing the same for the second walk we conclude

$$\|S_{k^*(\theta,n,2)}^{(2)}\| + \|S_{k^*(\theta,n,1)}^{(1)}\| \leq_{\delta,\varepsilon} M_{n,2}(\theta) + M_{n,1}(\theta).$$

Thus when bounding the sum, the previous results are sufficient from here.

We cannot do the same for the difference

$$\|S_{k^*(\theta,n,2)}^{(2)}\| - \|S_{k^*(\theta,n,1)}^{(1)}\|$$

as the bounds will then yield, for positive real numbers $a > b$

$$a\|2\| - b\|1\| = b(\|1\| - \|2\|) + (a - b)\|2\|$$

which is $O(n)$.

Instead,

$$\|S_{k^*(\theta,n,2)}^{(2)}\| - \|S_{k^*(\theta,n,1)}^{(1)}\| = (\|S_n^{(2)}\| - \|S_n^{(1)}\|) + (\|S_{k^*(\theta,n,2)}^{(2)}\| - \|S_n^{(2)}\|) - (\|S_{k^*(\theta,n,1)}^{(1)}\| - \|S_n^{(1)}\|)$$

Intuitively,

- the 1st part will be $O(\sqrt{n})$ as the drifts have the same magnitudes and these cancel each other out;
- the 2nd and 3rd part don't grow with n , they should be $O(1)$.

If there is a need to remove dependence on θ , consider the following. We'd expect the vertex with the maximal norm to “win”, but another vertex with a smaller norm but larger cosine could also win. Suppose we make the angle θ larger; both cosines will increase, but the smaller cosine will increase relatively more.² Therefore, the larger norms only gain more of an advantage as θ increases, so

$$\theta \mapsto \left\| S_{k^*(\theta, n, 1)}^{(1)} \right\|, \quad \theta \in [0, \varepsilon]$$

is increasing. With the analogous claim for the second walk,

$$\theta \mapsto \left\| S_{k^*(\theta, n, 2)}^{(2)} \right\| - \left\| S_{k^*(\theta, n, 1)}^{(1)} \right\|, \quad \theta \in [0, \varepsilon]$$

is maximal for $\theta = 0$.

To summarise this section:

- on $[0, \varepsilon]$, we use the above;
- elsewhere, the arguments are almost the same as in the previous section.

²Suppose $-\pi/2 < \alpha < \beta < -\varepsilon$; we claim that

$$\frac{\cos(\alpha + \varepsilon)}{\cos \alpha} > \frac{\cos(\beta + \varepsilon)}{\cos \beta}.$$

We know that the derivative of $\cos$ is decreasing, so

$$\frac{\cos(\alpha + \varepsilon) - \cos \alpha}{\varepsilon} > \frac{\cos(\beta + \varepsilon) - \cos \beta}{\varepsilon}.$$

We can use this in

$$\frac{\cos(\alpha + \varepsilon)}{\cos \alpha} - 1 = \frac{\cos(\alpha + \varepsilon) - \cos \alpha}{\cos \alpha} > \frac{\cos(\beta + \varepsilon) - \cos \beta}{\cos \alpha} > \frac{\cos(\beta + \varepsilon) - \cos \beta}{\cos \beta} = \frac{\cos(\beta + \varepsilon)}{\cos \beta} - 1.$$